

What is Mlflow and why we should consider working with it?

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MLSECOPS IN ACTION COURSE

We will work with Mlflow, because:

- We use multiple ML frameworks.
- We want an end-to-end ML platform.
- We prioritize simplicity and ease of use.
- We need comprehensive model management.
- We're a smaller team or just starting with model serving.



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به ساده‌ترین زبان ممکن، همان‌طور که گیت قابلیت‌ها
می‌دهد که کدهای خود را ورژن کنیم، ام‌ال‌فلو ما را قادر
خواهد ساخت که مدل‌های یادگیری ماشین خود را بصورت
استاندارد مدیریت و ورژن نمائیم.

MLflow is developed by...

MLflow is an open-source platform for the complete machine-learning lifecycle that allows you to manage your end-to-end machine-learning workflow. It was developed by **Databricks** and is now a Linux Foundation project.



Why MLflow matters?

The Chaos Without MLflow:

Imagine your team has 5 data scientists running hundreds of experiments. Without MLflow, you'd have notebooks scattered everywhere, models saved in random folders, and no one remembers which hyperparameters produced the best results from last month.

Key Benefits of MLflow

- 1. Experiment Tracking:** Automatically logs all your parameters, metrics, and artifacts so you never lose track of what worked. No more "I got 95% accuracy last week but can't remember how".
- 2. Model Versioning & Registry:** Centralized storage with automatic versioning - no more files named `model_final_v2_actually_final.pkl`. Clean progression from development to production.
- 3. Reproducibility:** Anyone can load and run your exact model months later with one line of code. Perfect for debugging or building on previous work.
- 4. Deployment Simplification:**

```
mlflow models serve -m "models:/MyModel/Production" -p 5000
```
- 5. Collaboration & Comparison:** Your team can compare 50 experiments in MLflow UI instantly, seeing which combination of features, algorithms, and hyperparameters worked best.

Key Components of MLflow



Core ML Lifecycle Features

Feature	Component	Description
Experiment Tracking	MLflow Tracking	Log and query experiments: code, data, config, and results
	Metrics Logging	Track metrics like accuracy, loss, F1-score over time
	Parameter Logging	Record hyperparameters and configuration settings
	Artifact Storage	Store model files, plots, datasets, and other outputs
	Run Comparison	Compare multiple experiment runs side-by-side

Core ML Lifecycle Features

Feature	Component	Description
Model Management	MLflow Models	Package ML models in multiple flavors (sklearn, tensorflow, etc.)
	Model Registry	Centralized model store with versioning and stage transitions
	Model Lineage	Track model ancestry and dependencies
Reproducibility	MLflow Projects	Package data science code in reusable, reproducible format
	Environment Management	Capture dependencies and execution environment
	Git Integration	Link runs to specific code commits

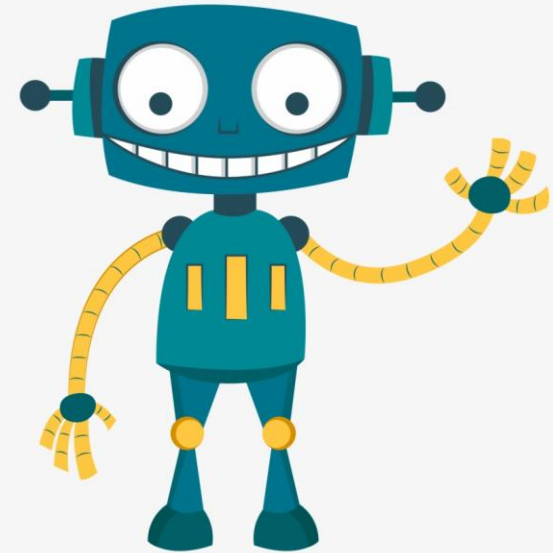
Deployment & Serving Features

Feature	Description
Multi-Platform Serving	Deploy to cloud platforms, Kubernetes, or local servers
Real-time Serving	Serve models via REST API endpoints
Batch Inference	Run models on large datasets
Model Serving	Deploy models via REST API or batch inference

Integration & Interface Features

Category	Feature	Description
Framework Support	ML Libraries	Works with scikit-learn, TensorFlow, PyTorch, R, Java, etc.
Storage Backends	Data Storage	Supports local files, S3, Azure Blob, Google Cloud Storage
	Database Support	MySQL, PostgreSQL, SQLite for metadata storage
User Interfaces	Web UI	Browser-based interface for exploring experiments and models
	API Access	Python, R, Java, and REST APIs
	Command Line	CLI tools for project management and deployment

mlflow™



Now Let's go for practical stuffs 🚀